HVAC PIPING NOTES

A. PROVIDE ALL MATERIALS AND EQUIPMENT AND PERFORM ALL LABOR REQUIRED TO INSTALL COMPLETE AND OPERABLE PIPING SYSTEMS AS INDICATED ON THE DRAWINGS, AS SPECIFIED AND AS REQUIRED BY CODE.

B. ELEVATIONS AS SHOWN ON THE DRAWINGS ARE TO THE CENTERLINE OF ALL PRESSURE PIPING AND TO THE INVERT OF ALL GRAVITY PIPING.

C. MAINTAIN A MINIMUM OF 3'6" OF GROUND COVER OVER ALL UNDERGROUND HVAC PIPING.

D. UNLESS OTHERWISE NOTED, ALL CHILLED WATER AND HEATING WATER PIPING SHALL BE 3⁄4 INCH SIZE.

E. PROVIDE AN AIR VENT AT THE HIGH POINT OF EACH DROP IN THE HEATING WATER, CHILLED WATER, AND OTHER CLOSED WATER PIPING SYSTEMS. ALL PIPING SHALL GRADE TO LOW POINTS. PROVIDE HOSE END DRAIN VALVES AT THE BOTTOM OF ALL RISERS AND LOW POINTS.

F. UNLESS OTHERWISE NOTED, ALL PIPING IS OVERHEAD, TIGHT TO UNDERSIDE OF STRUCTURE OR SLAB, WITH SPACE FOR INSULATION IF REQUIRED.

G. INSTALL PIPING SO THAT ALL VALVES, STRAINERS, UNIONS, TRAPS, FLANGES, AND OTHER APPURTENANCES REQUIRING ACCESS ARE ACCESSIBLE.

H. ALL VALVES SHALL BE INSTALLED SO THAT VALVE REMAINS IN SERVICE WHEN EQUIPMENT OR PIPING ON EQUIPMENT SIDE OF VALVE IS REMOVED.

I. ALL BALANCING VALVES AND BUTTERFLY VALVES SHALL BE PROVIDED WITH POSITION INDICATORS AND MAXIMUM ADJUSTABLE STOPS (MEMORY STOPS).

J. PROVIDE CHAINWHEEL OPERATORS FOR ALL VALVES IN EQUIPMENT ROOMS MOUNTED GREATER THAN 7 '-0" ABOVE FLOOR LEVEL; CHAIN SHALL EXTEND TO 7'-0" ABOVE FLOOR LEVEL.

K. ALL VALVES (EXCEPT CONTROL VALVES) AND STRAINERS SHALL BE FULL SIZE OF PIPE BEFORE REDUCING SIZE TO MAKE CONNECTIONS TO EQUIPMENT AND CONTROLS.

L. UNIONS AND/OR FLANGES SHALL BE INSTALLED AT EACH PIECE OF EQUIPMENT, IN BYPASSES, AND IN LONG PIPING RUNS (100 FEET OR MORE) TO PERMIT DISASSEMBLY FOR ALTERATION AND REPAIRS.

M. PITCH STEAM PIPING DOWNWARD IN THE DIRECTION OF FLOW 1-4 INCH IN 10 FEET (1 INCH IN 40 FEET) MINIMUM. PITCH ALL STEAM RETURN LINES DOWNWARD IN THE DIRECTION OF CONDENSATE FLOW 1⁄2 INCH PER 10 FEET (1 INCH IN 20 FEET) MINIMUM. WHERE LENGTH OF BRANCH LINES ARE LESS THAN 8 FEET, PITCH BRANCH LINES TOWARD MAINS 1⁄2 INCH PER FOOT MINIMUM.

N. PITCH UP ALL CONDENSATE RUNOUTS TO RISERS AND EQUIPMENT 1⁄2 INCH PER FOOT. WHERE THIS PITCH CANNOT BE OBTAINED, RUNOUTS OVER 8 FEET IN LENGTH SHALL BE ONE SIZE LARGER THAN NOTED.

O. INSTALL ALL PIPING WITHOUT FORCING OR SPRINGING. ALL PIPING SHALL CLEAR DOORS AND WINDOWS.

P. ALL VALVES SHALL BE ADJUSTED FOR SMOOTH AND EASY OPERATION.

Q. ALL PIPING WORK SHALL BE COORDINATED WITH ALL TRADES INVOLVED. OFFSETS IN PIPING AROUND OBSTRUCTIONS SHALL BE PROVIDED AT NO ADDITIONAL COST TO THE OWNER.

R. PROVIDE FLEXIBLE CONNECTIONS IN ALL PIPING SYSTEMS CONNECTED TO PUMPS, CHILLERS, COOLING TOWERS, AND OTHER EQUIPMENT WHICH REQUIRE VIBRATION ISOLATION EXCEPT WATER COILS. FLEXIBLE CONNECTIONS SHALL BE PROVIDED AS CLOSE TO THE EQUIPMENT AS POSSIBLE OR AS INDICATED ON THE DRAWINGS.

S. SLOPE REFRIGERANT PIPING ONE PERCENT IN THE DIRECTION OF OIL RETURN. LIQUID LINES MAY BE INSTALLED LEVEL.

T. INSTALL HORIZONTAL REFRIGERANT HOT GAS DISCHARGE PIPING WITH 1⁄2" PER 10 FEET DOWNWARD SLOPE AWAY FROM THE COMPRESSOR.

U. INSTALL HORIZONTAL REFRIGERANT SUCTION LINES WITH 1⁄2" PER 10 FEET DOWNWARD SLOPE TO THE COMPRESSOR, WITH NO LONG TRAPS OR DEAD ENDS WHICH MAY CAUSE OIL TO SEPARATE FROM THE SUCTION GAS AND RETURN TO THE COMPRESSOR IN DAMAGING SLUDGE.

V. PROVIDE LINE SIZE LIQUID INDICATORS IN MAIN LIQUID LINE LEAVING CONDENSER OR RECEIVER. INSTALL MOISTURE-LIQUID INDICATORS IN LIQUID LINES BETWEEN FILTER DRYERS AND THERMOSTATIC EXPANSION VALVES AND IN LIQUID LINE TO RECEIVER.

W. PROVIDE LINE SIZE STRAINER UPSTREAM OF EACH AUTOMATIC VALVE. PROVIDE SHUTOFF VALVE ON EACH SIDE OF STRAINER.

X. PROVIDE PERMANENT FILTER DRYERS IN LOW TEMPERATURE SYSTEMS AND SYSTEMS USING HERMETIC COMPRESSORS. PROVIDE EPLACEABLE CARTRIDGE FILTER DRYERS WITH THREE VALVE BYPASS ASSEMBLY FOR SOLENOID VALVES, ADJACENT TO RECEIVERS.

Y. PROVIDE REFRIGERANT CHARGING VALVE CONNECTIONS IN LIQUID LINE BETWEEN RECEIVER SHUTOFF VALVE AND EXPANSION VALVE.

DISCLAIMER NOTE: NOT ALL SPECIFICATIONS OR NOTES ON THIS SHEET MAY APPLY TO THIS SPECIFIC PROJECT. REFER TO PROJECT MECHANICAL PLANS FOR SCOPE OF WORK APPLICABLE TO THIS PROJECT.

MECHANICAL NOTES

A. CONTRACTOR SHALL FURNISH A WRITTEN GUARANTEE COVERING ALL WORKMANSHIP AND MATERIALS FOR A PERIOD OF ONE YEAR FROM DATE OF ACCEPTANCE.

B. CONTRACTOR SHALL PROVIDE POSITIVE MEANS FOR BALANCING EACH INDIVIDUAL AIR CONDITIONING SUPPLY AIR OUTLET, AS PER SCHEDULES & DRAWINGS, SYSTEMS SHALL BE BALANCED AGAINST THE ACTUAL INSTALLED STATIC PRESSURE.

C. BEFORE SUBMITTING A FINAL PROPOSAL, THE CONTRACTOR SHALL EXAMINE THE SITE OF THE PROPOSED WORK TO DETERMINE THE EXISTING CONDITIONS THAT MAY AFFECT HIS WORK AS HE WILL BE RESPONSIBLE FOR ANY ASSUMPTIONS MADE BY HIM IN REGARDS HERETO.

D. ANY APPARATUS, APPLIANCE, MATERIAL, WORK OR INCIDENTAL ACCESSORIES OR MINOR DETAILS NOT SHOWN BUT NECESSARY TO MAKE THE WORK COMPLETE AND PERFECT IN ALL RESPECTS AND READY FOR OPERATION, EVEN IF NOT PARTICULARLY SPECIFIED SHALL BE PROVIDED BY THE CONTRACTOR WITHOUT ADDITIONAL EXPENSE TO THE OWNER.

E. CONTRACTOR SHALL ASSUME RESPONSIBILITY FOR ALL COSTS INCURRED RESULTING FROM SUBSTITUTION OF EQUIPMENT AS WELL AS THE PERFORMANCE AND SPACE REQUIREMENTS OF SUCH EQUIPMENT.

F. ALL TESTS SHALL BE PERFORMED AS REQUIRED DURING THE DIFFERENT STAGES OF WORK AND A FINAL 24 HOUR MINIMUM RUNNING TEST SHALL BE DONE AFTER ALL OTHER TESTS AND BALANCING OPERATIONS HAVE BEEN DONE.

G. TEST AND BALANCE OF ALL EQUIPMENT SHALL BE PERFORMED BY AN INDEPENDENT TEST AND BALANCE COMPANY WITH A MINIMUM OF 5 YEARS EXPERIENCE IN PROJECTS OF THIS SIZE OR LARGER. THREE COPIES OF THE TEST AND BALANCE REPORT SHALL BE SUBMITTED TO THE ENGINEER FOR REVIEW AND APPROVAL.

H. CONTRACT DOCUMENTS FOR THE WORK ARE BASED ON AS-BUILT DRAWINGS PROVIDED BY OWNER. ANY DISCREPANCIES BETWEEN THE WORK DEPICTED IN THE DOCUMENTS AND THE ACTUAL CONDITIONS ENCOUNTERED SHALL BE BROUGHT PROMPTLY TO THE EOR FOR REVIEW AND CLARIFICATION.

I. ANY ADDITIONAL DUCTWORK REQUIRED TO MAKE FINAL EXHAUST DUCTWORK CONNECTIONS SHALL BE METAL PER FMC 501.4. CONSTRUCTION SHALL COMPLY WITH FMC 603. ANY DEFICIENCIES IN EXISTING DUCTWORK CONSTRUCTION SHALL BE REMOVED AND MADE COMPLIANT WITH THE ABOVE LISTED CODE SECTIONS.

J. ALL INDICATED AIR FLOWS ARE THE MINIMUM REQUIRED FOR THE ZONE. IF MORE AIR FLOW IS PROVIDED DISTRIBUTE THE ACCESS AIR PROPORTIONALLY, BY AREA, TO EACH ZONE SUPPLIED BY THE SYSTEM.

K. ALL RUNOUTS TO AIR TERMINALS TO BE SAME SIZE AS AIR TERMINAL CONNECTION UNLESS OTHERWISE NOTED.

L. PROVIDE VOLUME BALANCING DAMPERS FOR ALL DUCT BRANCHES CONNECTED TO AIR TERMINALS. ALL DUCTWORK TO BE SEALED SMACNA CLASS A.

M. T-STATS AND SENSORS TO BE MOUNTED 44" ABOVE FINISHED FLOOR, UNLESS OTHERWISE NOTED. PROVIDE .55° F DISCHARGE AIR TEMPERATURE FROM AIR HANDLING UNIT. ALL REQUIRED THERMOSTATS SHALL BE LOCATED AS SHOWN ON DRAWINGS OR WITHIN THE ZONE OF THE UNIT THEY ARE CONTROLLING, UNLESS OTHERWISE NOTED.

N. SUPPORT SUSPENDED MECHANICAL EQUIPMENT WITH ALL-THREAD RODS AND VIBRATION ISOLATORS.

O. PROVIDE GENERAL CONTRACTOR WITH FINAL SIZES AND LOCATIONS OF ALL WALL, FLOOR, AND ROOF PENETRATIONS TO COORDINATE ANY REQUIRED STRUCTURAL FRAMING MEMBERS.

P. ALL PENETRATIONS OF DUCTWORK, PIPING CONDUITS, AND VENTS THROUGH FIRE-RATED AND SMOKE-RATED BARRIERS SHALL HAVE FIRESTOP AND/OR SMOKE STOP PROTECTION IN ACCORDANCE WITH THE MINIMUM BUILDING CODE.

Q. IF AC UNITS MUST BE USED DURING CONSTRUCTION, THEY SHALL HAVE FILTRATION MEDIA WITH A MERV 8 AT EACH RETURN AIR GRILLE. THESE MUST BE REPLACED PRIOR TO OCCUPANCY.

R. HOLD ALL PIPING AND DUCTWORK TIGHT TO STRUCTURE IN ALL AREAS. NOTE THERE ARE HIGH CEILINGS IN MANY AREAS WHICH WILL REQUIRE CLOSE COORDINATION WITH OTHER TRADES DURING SHOP DRAWING PREPARATION AND DURING CONSTRUCTION.

S. SMOKE DETECTORS AND REMOTE ANNUNCIATOR ARE SUPPLIED BY MECHANICAL CONTRACTOR. SMOKE DETECTORS SHALL BE INSTALLED AND INTERLOCKED FOR SHUTDOWN IN ACCORDANCE WITH DIV. 23. SEE SPECIFICATIONS.

T. THREE SETS OF FILTERS ARE TO BE PROVIDED BY CONTRACTOR FOR EACH AC UNIT. ONE SET IS FOR USE DURING CONSTRUCTION, ONE SET TO BE INSTALLED PRIOR TO TEST AND BALANCE, AND A FINAL SET TO BE INSTALLED JUST PRIOR TO OCCUPANCY.

U. ALL MECHANICAL EQUIPMENT APPROACHING END OF USEFUL LIFE (15 YEARS) WILL BE REPLACED WITH LIKE SIZE EQUIPMENT. EFFICIENCIES SHALL BE COMPLIANT WITH LOCAL CODES.

V. CONTRACTOR SHALL BE RESPONSIBLE FOR INSTALLING ALL NECESSARY ACCESS PANELS TO PROVIDE ACCESS TO CONCEALED EQUIPMENT AND BALANCE DAMPERS, REGARDLESS OF WHETHER SAID ACCESS PANELS ARE SHOWN ON THE PLANS.

SHEET METAL NOTES

A. PROVIDE ALL MATERIALS AND EQUIPMENT AND PERFORM ALL LABOR REQUIRED TO INSTALL COMPLETE AND OPERABLE HVAC SYSTEMS AS INDICATED ON THE DRAWINGS, AS SPECIFIED AND AS REQUIRED BY CODE.

B. CERTAIN ITEMS SUCH AS RISES AND DROPS IN DUCTWORK, ACCESS DOORS, VOLUME DAMPERS, ETC., ARE INDICATED ON THE CONTRACT DOCUMENT DRAWINGS FOR CLARITY FOR A SPECIFIC LOCATION REQUIREMENT AND SHALL NOT BE INTERPRETED AS THE EXTENT OF THE REQUIREMENTS FOR THESE ITEMS.

C. IN CORRIDORS WHERE CEILING SPEAKERS AND AIR DIFFUSERS ARE INDICATED BETWEEN THE SAME LIGHT FIXTURES, INSTALL BOTH DEVICES AT THE QUARTER POINTS BETWEEN THE SAME FIXTURE.

D. UNLESS OTHERWISE SHOWN, LOCATE ALL ROOM THERMOSTATS AND HUMIDISTATS 4'-0" (CENTERLINE) ABOVE FINISHED FLOOR. NOTIFY THE ENGINEER OF ANY ROOMS WHERE THE ABOVE LOCATION CANNOT BE MAINTAINED OR WHERE THERE IS A QUESTION ON LOCATION.

E. ALL DUCTWORK SHALL CLEAR DOORS AND WINDOWS.

F. ALL DUCTWORK DIMENSIONS, AS SHOWN ON THE DRAWINGS, ARE INTERNAL CLEAR DIMENSIONS AND DUCT SIZE SHALL BE INCREASED TO COMPENSATE FOR DUCT LINING THICKNESS.

G. PROVIDE ALL 90 DEGREE SQUARE ELBOWS WITH DOUBLE RADIUS TURNING VANES UNLESS OTHERWISE INDICATED. ELBOWS IN DISHWASHER, KITCHEN, AND LAUNDRY EXHAUST SHALL BE UNVANED SMOOTH RADIUS CONSTRUCTION WITH A RADIUS EQUAL TO 1 1⁄2 TIMES THE WIDTH OF THE DUCT. PROVIDE ACCESS DOORS UPSTREAM OF ALL ELBOWS WITH TURNING VANES.

H. COORDINATE DIFFUSER, REGISTER, AND GRILLE LOCATIONS WITH ARCHITECTURAL REFLECTED CEILING PLANS, LIGHTING, AND OTHER CEILING ITEMS AND MAKE MINOR DUCT MODIFICATIONS TO SUIT.

I. FIELD ERECTED AND FACTORY ASSEMBLED AIR HANDLING UNIT COILS SHALL BE ARRANGED FOR REMOVAL FROM THE UPSTREAM SIDE WITHOUT DISMANTLING SUPPORTS. PROVIDE GALVANIZED STRUCTURAL STEEL SUPPORTS FOR ALL COILS (EXCEPT LOWEST COIL) IN BANKS OVER TWO COILS HIGH TO PERMIT INDEPENDENT REMOVAL OF ANY COIL.

J. ALL AC UNITS SHALL OPERATE WITHOUT MOISTURE CARRYOVER.

K. LOCATE ALL MECHANICAL EQUIPMENT (SINGLE DUCT, DUAL DUCT, VARIABLE VOLUME, CONSTANT VOLUME AND FAN POWERED BOXES, FAN COIL UNITS, CABINET HEATERS, UNIT HEATERS, UNIT VENTILATORS, COILS, STEAM HUMIDIFIERS, ETC.) FOR UNOBSTRUCTED ACCESS TO UNIT ACCESS PANELS, CONTROLS AND VALVING.

L. FINNED TUBE RADIATION ENCLOSURES SHALL BE WALL TO WALL UNLESS OTHERWISE INDICATED.

M. PROVIDE FLEXIBLE CONNECTIONS IN ALL DUCTWORK SYSTEMS (SUPPLY, RETURN, AND EXHAUST) CONNECTED TO AIR HANDLING UNITS, FANS, AND OTHER EQUIPMENT WHICH REQUIRE VIBRATION ISOLATION. FLEXIBLE CONNECTIONS SHALL BE PROVIDED AT THE POINT OF CONNECTION TO THE EQUIPMENT UNLESS OTHERWISE INDICATED.

N. UNLESS OTHERWISE NOTED, ALL DUCTWORK IS OVERHEAD, TIGHT TO THE UNDERSIDE OF THE STRUCTURE, WITH SPACE FOR INSULATION IF REQUIRED.

O. RUNS OF FLEXIBLE DUCT SHALL NOT EXCEED 5 FEET (EDIT MAXIMUM LENGTH OF FLEXIBLE DUCT TO SUIT PROJECT; 5 FEET MAXIMUM RECOMMENDED LENGTH, 8 FEET MAXIMUM LENGTH).

P. ALL DUCTWORK SHALL BE COORDINATED WITH ALL TRADES INVOLVED. OFFSETS IN DUCTS, INCLUDING DIVIDED DUCTS AND TRANSITIONS AROUND OBSTRUCTIONS, SHALL BE PROVIDED AT NO ADDITIONAL COST TO THE OWNER.

Q. PROVIDE ACCESS DOORS IN DUCTWORK TO PROVIDE ACCESS FOR ALL SMOKE DETECTORS, FIRE DAMPERS, SMOKE DAMPERS, VOLUME DAMPERS, HUMIDIFIERS, COILS, AND OTHER ITEMS LOCATED IN THE DUCTWORK WHICH REQUIRE SERVICE AND/OR INSPECTION.

R. PROVIDE ACCESS DOORS IN DUCTWORK FOR OPERATION, ADJUSTMENT, AND MAINTENANCE OF ALL FANS, VALVES, AND MECHANICAL EQUIPMENT.

S. ALL DUCTS SHALL BE GROUNDED ACROSS FLEXIBLE CONNECTIONS WITH FLEXIBLE COPPER GROUNDING STRAPS. GROUNDING STRAPS SHALL BE BOLTED OR SOLDERED TO BOTH THE EQUIPMENT AND THE DUCT.

T. SMOKE DETECTORS SHALL BE FURNISHED AND WIRED BY THE ELECTRICAL CONTRACTOR. THE MECHANICAL CONTRACTOR SHALL BE RESPONSIBLE FOR MOUNTING THE SMOKE DETECTOR IN DUCTWORK AS SHOWN ON THE DRAWINGS AND IN ACCORDANCE WITH MANUFACTURER'S PRINTED INSTRUCTIONS.

U. TERMINATE GAS VENTS FOR UNIT HEATERS, WATER HEATERS, HIGH PRESSURE PARTS WASHER, HIGH PRESSURE CLEANER, AND OTHER GAS APPLIANCES A MINIMUM OF 3'0" ABOVE ROOF WITH RAIN CAP (EDIT APPLIANCES AND HEIGHT ABOVE ROOF TO MEET CODE AND TO SUIT PROJECT REQUIREMENTS).

V. SEE SPECIFICATIONS FOR DUCTWORK GAUGES, BRACING, HANGERS, AND OTHER REQUIREMENTS. EXTERIOR LOUVERS ARE INDICATED FOR INFORMATION ONLY. DETAILED DESCRIPTIONS ARE PROVIDED IN THE ARCHITECTURAL SPECIFICATIONS.

W. EXTERIOR LOUVERS ARE INDICATED FOR INFORMATION ONLY. LOUVER SIZES, LOCATIONS, AND DETAILS SHALL BE COORDINATED WITH GENERAL CONTRACTOR.

X. EXTERIOR LOUVERS ARE INDICATED FOR INFORMATION ONLY. LOUVER SIZES, LOCATIONS, MOUNTING, AND DETAILS SHALL BE COORDINATED WITH OTHER TRADES INVOLVED.

ELECTRICAL REQUIRMENTS FOR MECHANICAL EQUIPMENT

GENERAL REQUIREMENTS

A. DIVISION 23 CONTRACTOR SHALL PROVIDE MECHANICAL EQUIPMENT WITH VOLTAGES AND OTHER ELECTRICAL CHARACTERISTICS AS INDICATED ON THE DRAWINGS AND WITHIN THE SPECIFICATIONS.

B. ALL STARTERS, DISCONNECT SWITCHES, MOTOR CONTROL CENTERS, AND VARIABLE FREQUENCY DRIVES, FOR EQUIPMENT PROVIDED UNDER DIVISION 23, SHALL BE FURNISHED UNDER DIVISION 23 AND INSTALLED UNDER DIVISION 26. FACTORY MOUNTED STARTERS, DISCONNECTS SWITCHES, AND VARIABLE FREQUENCY DRIVES SHALL BE FURNISHED AND INSTALLED UNDER DIVISION 23. HVAC CONTRACTOR SHALL PROVIDE STARTERS, DISCONNECT SWITCHES, MOTOR CONTROL CENTERS AND VARIABLE FREQUENCY DRIVES IN ACCORDANCE WITH DIVISION 26.

C. DIVISION 23 SHALL SUBMIT WIRING DIAGRAMS TO THE ARCHITECT/ENGINEER FOR APPROVAL AND PROVIDE APPROVED DIAGRAMS TO THE DIVISION 26 CONTRACTOR SO THAT THE ELECTRICAL WORK MAY BE PROPERLY ACCOMPLISHED.

D. DIVISION 23 CONTRACTOR SHALL FULLY COOPERATE WITH THE OTHER DIVISIONS AND TRADES ON THE PROJECT, AND THEIR MANUFACTURERS, IN PROMPTLY PROVIDING THE INFORMATION REQUIRED FOR PROPER COORDINATION OF MOTOR PROTECTION, CONTROL EQUIPMENT AND WIRING, AND THE OTHER CHARACTERISTICS OF THE EQUIPMENT.

E. IT SHALL BE THE RESPONSIBILITY OF THE DIVISION 23 AND DIVISION 26 CONTRACTORS TO CHECK FOR ADEQUACY OF SUPPLY WIRING, OVERCURRENT PROTECTION, PROPER VOLTAGE, PHASE ROTATION AND FINAL LOCATION OF EQUIPMENT PROVIDED PRIOR TO THE RUNNING OF ANY CONDUIT OR WIRING. COORDINATE WITH DIVISION 26 TO ASSURE PROPER ELECTRICAL SERVICE IS PROVIDED TO EQUIPMENT UNDER DIVISION 23.

F. EQUIPMENT CONNECTIONS SHALL BE MADE THROUGH CONDUIT OR RACEWAYS IN ACCORDANCE WITH DIVISION 26, WITH THE EXCEPTION THAT CONNECTIONS TO MOTORS SHALL BE MADE THROUGH LIQUID TIGHT FLEXIBLE METAL CONDUIT WITH EQUIPMENT GROUNDING CONDUCTOR.

POWER REQUIREMENTS

A. ALL POWER WIRING FOR MECHANICAL EQUIPMENT PROVIDED UNDER DIVISION 23 SHALL BE FURNISHED AND INSTALLED UNDER DIVISION 26 TO THE POINT OF FINAL CONNECTION (FROM THE SOURCE TO THE STARTERS, DISCONNECT SWITCHES, MCC'S OR VFD'S AND TO EQUIPMENT, MOTOR, OR OTHER CONNECTION POINT). EXCEPT WHERE SPECIFICALLY INDICATED OTHERWISE, ALL POWER WIRING TO THE POINT OF FINAL CONNECTION FOR EQUIPMENT PROVIDED UNDER DIVISION 23 SHALL BE ACCOMPLISHED UNDER DIVISION 26.

B. FINAL ELECTRICAL POWER CONNECTIONS TO ALL EQUIPMENT SHALL BE FURNISHED AND INSTALLED UNDER DIVISION 26. IN GENERAL, THE POINT OF FINAL CONNECTION SHALL BE THE TERMINAL HOUSING ON THE EQUIPMENT, MOTOR, OR AN INTERNAL JUNCTION BOX ON THE EQUIPMENT OR ITEM. IF NO JUNCTION BOX IS FURNISHED, A JUNCTION BOX SHALL BE FURNISHED AND INSTALLED UNDER DIVISION 26. WIRE LEADS F ADEQUATE LENGTH TO ENSURE A PROPER CONNECTION AT THE FINAL LOCATION SHALL BE FURNISHED AND INSTALLED UNDER DIVISION 26.

C. WIRING, OVERCURRENT PROTECTION DEVICES, VOLTAGE, PHASE, ROTATION AND FINAL LOCATION OF ALL EQUIPMENT PROVIDED UNDER DIVISION 23 SHALL BE COORDINATED WITH ALL SIMILAR DEVICES AND POWER WIRING URNISHED AND INSTALLED UNDER DIVISION 26. COORDINATION SHALL BE ACCOMPLISHED PRIOR TO THE RUNNING OF ANY CONDUIT OR WIRING.

CONTROL REQUIREMENTS

A. ALL CONTROL WIRING (LINE VOLTAGE AND/OR LOW VOLTAGE) FOR MECHANICAL EQUIPMENT PROVIDED UNDER DIVISION 23, SHALL BE FURNISHED AND INSTALLED UNDER DIVISION 23. WIRING FROM THE POWER SOURCE TO ALL CONTROL PANELS, DDC CONTROL PANELS, STARTERS, OPERATOR WORK STATIONS (PC'S, PRINTERS, MONITORS, AND OTHER WORK STATION EQUIPMENT) AND OTHER CONTROL EQUIPMENT REQUIRED FOR A COMPLETE AND OPERABLE CONTROL SYSTEM SERVING MECHANICAL EQUIPMENT PROVIDED UNDER DIVISION 23 SHALL BE FURNISHED AND INSTALLED UNDER DIVISION 23.

B. ALL WIRING FROM CONTROL PANELS TO CONTROL DEVICES FOR MECHANICAL EQUIPMENT PROVIDED UNDER DIVISION 23 SHALL BE FURNISHED AND INSTALLED UNDER DIVISION 23. ALL CONTROL WIRE AND CONDUIT SHALL COMPLY WITH THE NATIONAL ELECTRIC CODE AND DIVISION 26 OF THE SPECIFICATIONS. ALL CONTROL WIRING SHALL BE IN CONDUIT.

C. UNLESS OTHERWISE NOTED, ALL TEMPERATURE CONTROL DEVICES, INSTALLED PILOT DEVICES, AND ALL ASSOCIATED PILOT CONTROL EQUIPMENT, FOR ALL EQUIPMENT PROVIDED UNDER DIVISION 23, SHALL BE FURNISHED AND INSTALLED UNDER DIVISION 23.

D. ELECTRICAL CONTROL WIRING FOR CONNECTION OF TEMPERATURE CONTROLLERS, PUSH BUTTONS, INTERLOCKS IN MOTOR CONTROLLERS, PNEUMATIC SWITCHES AND LIKE ITEMS IS SPECIFIED UNDER THE CONTROL SECTION(S) IN DIVISION 23 AND INSTALLED BY DIVISION 23.

E. ALL MOTORS, MOUNTS, REMOTE MOUNTED PUSH BUTTON CONTROLS AND ALL SPEED CONTROL SWITCHES FOR MULTI- SPEED MOTORS FOR ALL EQUIPMENT PROVIDED UNDER DIVISION 23 SHALL BE FURNISHED AND INSTALLED UNDER DIVISION 23.

HVAC SYMBOLS:

- T — THERMOSTAT CONTROL
- SAT — DUCT MOUNTED SUPPLY AIR TEMPERATURE SENSOR
- RAT — DUCT MOUNTED RETURN AIR TEMPERATURE SENSOR
- AT — AVERAGING TEMPERATURE SENSOR
- HS — HUMIDITY SENSOR
- SD — SMOKE DETECTOR
- C — CARBON DIOXIDE (CO2) SENSOR
- U — 3/4" DOOR UNDERCUT
- G — OPEN VAULT DOOR WITH BARS
- FT — REMOTE FIRE TESTING STATION
- V — VOC SENSOR IN SUPPLY DUCT. EQUAL TO BUILDING AUTOMATION PRODUCTS, INC (BAPI). COORDINATE WITH IOT.
- PC — PARTICLE COUNTER IN RETURN DUCT. EQUAL TO SENSIRION AG SP530. COORDINATE WITH IOT.
- TR — TEMPERATURE AND RELATIVE HUMIDITY DISPLAY EQUAL TO AUTOMATION COMPONENTS, INC. COORDINATE WITH IOT.

HVAC/PIPING SYMBOLS:

- PIPE ELBOW UP
- PIPE ELBOW DOWN
- UNION (FLANGED CONNECTION/JOINT)
- END CAP
- TOP CONNECTION
- BOTTOM CONNECTION
- DIRECTION OF FLOW
- DIRECTION OF PITCH
- D — CONDENSATE DRAIN PIPING
- RL — REFRIGERANT LIQUID PIPING
- RS — REFRIGERANT SUCTION PIPING
- CLEAN OUT PIPING

HVAC LEGEND DOUBLE LINE:

- 12X6 — INTERIOR DUCT DIMENSIONS (W X D)
- DUCT TURNED UP (RETURN)
- DUCT TURNED DOWN (SUPPLY)
- R — RISE IN ELEVATION DIRECTION OF AIRFLOW
- D — DROP IN ELEVATION DIRECTION OF AIRFLOW
- DUCT TURN WITH RADIUS ELBOW
- SQUARE TURN WITH VANE ELBOW
- FLEXIBLE CONNECTION
- DUCT WITH INTERNAL LINING
- TEE-BOOT ENTRY BRANCH WITH MANUAL VOLUME CONTROL DAMPER
- TEE-ROUND ENTRY BRANCH WITH MANUAL VOLUME CONTROL DAMPER
- MOD — MOTOR OPERATED DAMPER
- AP — ACCESS PANELS
- SUPPLY AIR DUCT SECTION
- RETURN AIR DUCT SECTION
- OUTSIDE AIR DUCT SECTION
- EXHAUST AIR DUCT SECTION
- FLEXIBLE AIR DUCT
- RECTANGULAR TO ROUND DUCT TRANSITION
- FD — FIRE DAMPER IN HORIZONTAL DUCT
- FD — FIRE DAMPER IN VERTICAL DUCT
- FD / FSD — COMBINATION FIRE AND SMOKE DAMPER
- DIFFUSER WITH SECTORIZING BAFFLE (EXAMPLE: THREE WAY SUPPLY DIFFUSER)

HVAC EQUIPMENT INSTALLATION

A. INCLUDE ALL CODE AND CONSTRUCTION RELATED PERMITS.

B. INSTALLATION TO MEET ALL FEDERAL AND LOCAL CODE REQUIREMENTS.

C. RTU CONDENSATE LINES SHALL BE EXTENDED TO ROOF DRAINS, SCUPPERS, OR GUTTERS AND SHALL HAVE A CORRECT TRAP DESIGN.

D. RTUS SHALL BE PROPERLY SEATED AND SEALED ON THE ROOF CURB.

E. THERMOSTAT WIRES SHALL NOT BE SPLICED OUTSIDE OF THE HVAC UNIT. SHALL BE INSTALLED IN CONDUIT SEPARATELY FROM HIGH VOLTAGE AND SHALL NOTE BE INSIDE THE DUCTWORK.

F. FOLLOW OEM INSTRUCTIONS FOR STARTUP AND OPERATIONAL COMMISSIONING.

G. ALL ROOF PENETRATIONS SHALL BE SEALED WATER TIGHT.

H. MECHANICAL CONTRACTOR AND HVAC SYSTEM SUPPLIER SHALL CONFIRM WITH CONTROLS VENDOR TO ENSURE FULL COMPATIBILITY FOR ALL HVAC EQUIPMENT PRIOR TO ORDERING/PURCHASE.

I. MC, EC, AND CONTROL VENDOR TO ATTEND A CONTROLS INSTALLATION / DESIGN MEETING.

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Rev	Date	Comments
	07/02/2026	PERMIT SET

Client:

FIFTH THIRD BANK

Project Number: 26032
Project Name:

FIFTH THIRD BANK WATSON
1,900 SF

2711 WATSON BOULEVARD
WARNER ROBBINS, GA 31093

Key Plan:

Sheet Title:

MECHANICAL NOTES AND LEGEND

Sheet Number:

M-002

6/30/2026 4:54:50 PM